MATH102BKK · HARBOUR.SPACE BANGKOK

Sets & set operations

Tools for describing collections — and for proving when two collections are equal.

Session	Date	Today
2 of 15	Tue 8 Sep 2026	Sets, operations, proof by double inclusion

RECAP

Where we left off

- → Rules come with conditions; check them before using a rule.
- → For inequalities, the sign tells you whether the direction stays or reverses.
- → One counterexample is enough to show an “always” claim is false.
- → Marks live in the justification, not the answer.

TODAY'S QUESTION

Yesterday we argued about *numbers*. But "the solution set of $|2x-3|<5$" is not a number — it is a **collection**.

How do you state, and *prove*, that two collections are the same?

OBJECTIVES

By the end of today you can...

- → Use **roster** and **set-builder** notation, and distinguish in from $subseteq$
- → Compute union, intersection, difference, complement, symmetric difference and Cartesian product
- → Prove set equalities by **double inclusion** and by element-chasing
- → Compute power sets and state why $|\mathcal{P}(A)| = 2^{|A|}$

> **MAIN SKILL** The third. The other objectives give us the vocabulary; double inclusion is the proof method we will practise carefully.

A set is a collection, and nothing more

DEFINITION

A **set** is an unordered collection of distinct objects, called its **elements**.

$x \in A$ means "x is an element of A"; $x \notin A$ means it is not.

TWO NOTATIONS

Roster — list them: $A = \{1, 2, 3\}$

Set-builder — state the rule:

$$B = \{\, x \in \mathbb{Z} : x^2 < 10 \,\}$$

read "the x in $\mathbb{Z}$ such that $x^2 < 10$", so $B = \{-3, -2, -1, 0, 1, 2, 3\}$.

UNORDERED, AND NO REPEATS $\{1, 2, 2, 3\}$ and $\{3, 1, 2\}$ are the **same set** as $\{1, 2, 3\}$. A set records *what* is in it, never how often or in what order.

The set with nothing in it

DEFINITION The **empty set** $\varnothing = \{\ \}$ has no elements: $x \in \varnothing$ is false for every x.

IT SHOWS UP ON ITS OWN $\{\, x \in \mathbb{R} : x^2 < 0 \,\} = \varnothing$. Nobody chose this; the condition excludes everything.

TRAP · THREE DIFFERENT THINGS

$\varnothing$ — a set with no elements. $|\varnothing| = 0$.

$\{\varnothing\}$ — a set with **one** element, and that element is $\varnothing$. $|\{\varnothing\}| = 1$.

$\{0\}$ — a set with one element, the number zero. Not the same as either.

Element of, or subset of?

DEFINITION $A \subseteq B$ ("A is a **subset** of B") means: *every* element of A is an element of B.

$$A \subseteq B \iff \forall x\,(x \in A \Rightarrow x \in B)$$

THE DIFFERENCE IN ONE LINE $\in$ relates an **object** to a set. $\subseteq$ relates a **set** to a set.

EXAMPLE · $C = \{\{1\}, 2\}$

- $\{1\} \in C$ – true, it is listed.
- $\{1\} \subseteq C$ – **false**: is $1 \in C$? No.
- $1 \in C$ – false. C contains $\{1\}$, not 1.
- $\{2\} \subseteq C$ – true, since $2 \in C$.

Why $\varnothing \subseteq A$ for every set A

PROOF

To show $\varnothing \subseteq A$, we would need to find an element of $\varnothing$ that is not in A.

But $\varnothing$ has no elements. There is nothing to find.

So no element can break the subset condition, and $\varnothing \subseteq A$ for every set A.

$\blacksquare$

WHY IT MATTERS This is why $\varnothing$ appears in every power set: it is a subset of every set.

TRAP $\varnothing \subseteq A$ always. But $\varnothing \in A$ only if you *put* it there.

Two sets are equal when each contains the other

DEFINITION · SET EQUALITY

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Equivalently: A and B have exactly the same elements.

THE METHOD: DOUBLE INCLUSION

To prove $A = B$, write two proofs:

1. Take an arbitrary $x \in A$. Show $x \in B$. That gives $A \subseteq B$.
2. Take an arbitrary $x \in B$. Show $x \in A$. That gives $B \subseteq A$.

WHY THIS IS THE WHOLE SESSION The word arbitrary is what makes it a proof: you assumed nothing about x, so it holds for all of them.

ACTIVITY 1

True or false — and read it from the definition

ACTIVITY 1 PAIRS 10 MIN

Let $A = \{1, 2, \{3\}, \varnothing\}$. Decide true or false, and write the **one line of definition** that settles each.

1. $3 \in A$
2. $\{3\} \in A$
3. $\{3\} \subseteq A$
4. $\varnothing \in A$
5. $\varnothing \subseteq A$
6. $\{1, 2\} \subseteq A$
7. $|A| = 4$

Share: each pair takes one line and reads their justification to the room. Where two pairs disagree, settle it from the definition, not by vote.

Read the symbol, then read the definition

CLAIM		WHY
1 · $3 \in A$	False	A lists $\{3\}$, not 3.
2 · $\{3\} \in A$	True	It is listed.
3 · $\{3\} \subseteq A$	False	Would need $3 \in A$.
4 · $\varnothing \in A$	True	Only because it was listed.
5 · $\varnothing \subseteq A$	True	Vacuously, for *every* set.
6 · $\{1, 2\} \subseteq A$	True	Both are in A.
7 · $\lvert A \rvert = 4$	True	1, 2, $\{3\}$, $\varnothing$.

EXAMPLE · ROWS 2 AND 3 $\{3\} \in A$ asks if $\{3\}$ is *on the list*: yes. $\{3\} \subseteq A$ asks if its *contents* are: no.

Building new sets from old ones

UNION

$$A \cup B = \{\, x : x \in A \text{ or } x \in B \,\}$$

"or" is **inclusive** — x may be in both.

INTERSECTION

$$A \cap B = \{\, x : x \in A \text{ and } x \in B \,\}$$

Empty intersection ⇒ **disjoint**.

DIFFERENCE

$$A \setminus B = \{\, x : x \in A \text{ and } x \notin B \,\}$$

In A, not in B. Not symmetric.

WORKED · $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$ $A \cup B = \{1, 2, 3, 4, 5\} \cdot A \cap B = \{3, 4\} \cdot A \setminus B = \{1, 2\} \cdot B \setminus A = \{5\}$

Complement and symmetric difference

COMPLEMENT

Fix a universe U. For $A \subseteq U$:

$$A^c = U \setminus A = \{\, x \in U : x \notin A \,\}$$

SYMMETRIC DIFFERENCE

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

In exactly one of them — the "exclusive or" of sets.

TRAP You must state U before using A^c. In $\mathbb{Z}$, the complement of the evens is the odds. In $\mathbb{R}$, it also contains non-integers such as $\frac{1}{2}$.

WORKED With $U = \{1, \ldots, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$: $A^c = \{5, 6\}$, and $A \triangle B = \{1, 2\} \cup \{5\} = \{1, 2, 5\}$.

What a Venn diagram shows you

DEFINITION · VENN DIAGRAM A Venn diagram draws sets as overlapping regions — one region for each way an element can be in or out of each set.

EXAMPLE · TWO CIRCLES, FOUR REGIONS In both, in A only, in B only, in neither — that is $A \cap B$, $A \setminus B$, $B \setminus A$ and $(A \cup B)^c$. Every element of U lands in exactly one.

…and why it is not a proof

TRAP · A REAL EXAM TRAP

One picture shows *one* configuration. A claim about **all** sets is not settled by one picture.

Three sets need $2^3 = 8$ regions, and a hand-drawn diagram routinely omits some. With four, no diagram of circles can show all 16.

THE HONEST WORKFLOW

1. Draw the diagram to *guess* the identity.
2. Write the double-inclusion proof to *establish* it.
3. Hand in step 2.

ACTIVITY 2

Translate: prose into notation, notation into prose

ACTIVITY 2 GROUPS OF THREE 12 MIN

Universe $U = \{1, 2, \ldots, 10\},\ E = \{\text{even numbers in } U\},\ T = \{\text{multiples of 3 in } U\}$.

1. Write E and T in roster notation, then in set-builder notation.
2. Compute $E \cap T,\ E \cup T,\ E \setminus T,\ E \triangle T,\ T^c$.
3. Turn into notation: "the numbers in U that are even but not multiples of three".
4. Turn into plain English: $(E \cup T)^c$.

Share: one group reads their sentence for (4) aloud; the room checks it against the roster list they computed.

Both directions of the translation

1–2 · THE SETS, AND THE OPERATIONS

$E = \{2, 4, 6, 8, 10\},\ T = \{3, 6, 9\}$.

$E \cap T = \{6\}$ · $E \cup T = \{2, 3, 4, 6, 8, 9, 10\}$
$E \setminus T = \{2, 4, 8, 10\}$ · $E \triangle T = \{2, 3, 4, 8, 9, 10\}$
$T^c = \{1, 2, 4, 5, 7, 8, 10\}$

3 · PROSE INTO NOTATION "Even but not a multiple of three" is $E \setminus T = \{2, 4, 8, 10\}$.

4 · NOTATION INTO PROSE

$(E \cup T)^c$ is "neither even nor a multiple of three" — that is $\{1, 5, 7\}$.

Notice: *not (even or mult. of 3)* became *not even and not mult. of 3*. Hold that thought.

PART II

Proving that two sets are equal

Now use the vocabulary: prove a set equality one element at a time, or show a false identity with one clear example.

2

Proving an inclusion, one element at a time

THE METHOD

To show $X \subseteq Y$: take an **arbitrary** $x \in X$, unfold the definitions of the operations in X, and deduce $x \in Y$.

Each operation turns into a logical connective: $\cup \to$ or, $\cap \to$ and, $\setminus \to$ and-not.

THE SKELETON TO COPY

($\subseteq$) Let $x \in X$ be arbitrary. Then *[unfold]*, so *[deduce]*, hence $x \in Y$.

($\supseteq$) Let $x \in Y$ be arbitrary. Then *[unfold]*, so *[deduce]*, hence $x \in X$.

Both inclusions hold, so $X = Y$. $\blacksquare$

$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

($\subseteq$) — LEFT INTO RIGHT

Let $x \in A \setminus (B \cap C)$ be arbitrary.

Then $x \in A$ and $x \notin B \cap C$.

Now $x \notin B \cap C$ means **not** ($x \in B$ and $x \in C$) — so $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in A \setminus B$; if $x \notin C$, then $x \in A \setminus C$. Either way x lies in the union. ∎

THE STEP TO WATCH Negating "and" produced "or". That single move is De Morgan — two slides away, with a name.

...and the inclusion back again

($\supseteq$) – RIGHT INTO LEFT

Let $x \in (A \setminus B) \cup (A \setminus C)$ be arbitrary, so x is in at least one part.

Case 1: $x \in A \setminus B$. Then $x \in A$, $x \notin B$, so $x \notin B \cap C$.

Case 2: $x \in A \setminus C$. Then $x \in A$, $x \notin C$, so $x \notin B \cap C$.

Both give $x \in A \setminus (B \cap C)$. With the first half: equal. $\blacksquare$

WHY TWO CASES "Or" hands you one of two situations, not a specific one. Handle each, and both must land in the same place.

CONCEPT II · THE NAMED THEOREM

De Morgan's laws

THEOREM · DE MORGAN'S LAWS For all $A, B \subseteq U$:

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

PROOF OF THE FIRST

($\subseteq$) Let $x \in (A \cup B)^c$: then not ($x \in A$ or $x \in B$), so $x \notin A$ and $x \notin B$, i.e. $x \in A^c \cap B^c$.

($\supseteq$) Let $x \in A^c \cap B^c$: then x is in neither, so $x \notin A \cup B$. $\blacksquare$

IN ENGLISH "Not (even or a multiple of three)" $=$ "not even and not a multiple of three" — your Activity 2 sentence. Negation swaps *and* with *or*.

Showing a set identity is false

WHAT A DISPROOF NEEDS A set identity claims to work for **all** sets. To show it is false, give *one* example where the two sides differ.

WORKED · IS $A \setminus (B \setminus C) = (A \setminus B) \setminus C$?

Take $A = B = C = \{1\}$.

Left: $B \setminus C = \varnothing$, so $A \setminus \varnothing = \{1\}$.

Right: $A \setminus B = \varnothing$, so $\varnothing \setminus C = \varnothing$.
$\{1\} \neq \varnothing$: false.

TRAP

"It failed when I drew it" is not a disproof.

Name the sets, compute both sides, point at the element on one side and not the other — here, 1.

ACTIVITY 3

Prove it or break it

ACTIVITY 3 GROUPS OF THREE 15 MIN

Three are true and two are false. Prove a true one by double inclusion. For a false one, name the sets and compute both sides.

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $(A \setminus B) \cup B = A \cup B$
3. $(A \cup B) \setminus B = A$
4. $A \triangle A = \varnothing$
5. $A \cup (B \cap C) = (A \cup B) \cap C$

Share: two groups present, one proof and one disproof. The room checks that the proof says "arbitrary" and the disproof names its element.

Three survive, two do not

IDENTITY		PROOF SKETCH, OR THE WITNESS
1 · $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	True	Distributivity: the "or" splits into two cases.
2 · $(A \setminus B) \cup B = A \cup B$	True	Remove B, put it back.
3 · $(A \cup B) \setminus B = A$	False	$A = B = \{1\}$: $\varnothing$ against $\{1\}$.
4 · $A \triangle A = \varnothing$	True	Nothing is in A but not in A.
5 · $A \cup (B \cap C) = (A \cup B) \cap C$	False	$A = \{1\}$, $B = C = \varnothing$: $\{1\}$ against $\varnothing$.

REPAIRING 3 It holds exactly when $A \cap B = \varnothing$.

The Cartesian product

DEFINITION · CARTESIAN PRODUCT The Cartesian product of A and B is

$$A \times B = \{\, (a,b) : a \in A \text{ and } b \in B \,\}$$

Its elements are ordered pairs: $(a,b) = (c,d)$ exactly when $a = c$ and $b = d$.

ORDERED, UNLIKE A SET $\{1,2\} = \{2,1\}$, but $(1,2) \neq (2,1)$ – the difference that makes coordinates, and tomorrow's functions, possible.

WORKED · $A = \{1,2\}$, $B = \{X,Y\}$

$$A \times B = \{(1,x),(1,y),(2,x),(2,y)\}$$

$$B \times A = \{(x,1),(x,2),(y,1),(y,2)\}$$

TRAP $A \times B \neq B \times A$ in general. Witness: $A = \{1\}$, $B = \{2\}$.

How big is $A \times B$?

THEOREM

If A and B are finite, then $|A \times B| = |A| \cdot |B|$.

PROOF Every pair (a, b) comes from two independent choices: $|A|$ ways to pick a, then $|B|$ ways to pick b. Distinct choices give distinct pairs, since pairs match only when both coordinates do. So the count is $|A| \cdot |B|$. ■

YOU HAVE JUST USED THE PRODUCT RULE That argument — independent choices multiply — is the foundation of Session 7's combinatorics. Notice it now, so it is familiar when it gets a name.

The power set

DEFINITION · POWER SET The **power set** of A is the set of all subsets of A:

$$\mathcal{P}(A) = \{\, X : X \subseteq A \,\}$$

Its elements are themselves sets.

TRAP Members of $\mathcal{P}(A)$ are related to A by $\subseteq$, but to $\mathcal{P}(A)$ by $\in$. So $\{1\} \in \mathcal{P}(A)$ and $\{1\} \subseteq A$ — both true, different statements.

WORKED · $\mathcal{P}(\{1,2,3\})$

Organised by size:

$\varnothing$

$\{1\},\ \{2\},\ \{3\}$

$\{1,2\},\ \{1,3\},\ \{2,3\}$

$\{1,2,3\}$

Eight subsets: $1+3+3+1=8=2^3$. Both $\varnothing$ and A itself are always in there.

$|\mathcal{P}(A)| = 2^{|A|}$

THEOREM If $|A| = n$ then $\mathcal{P}(A)$ has exactly 2^n elements.

PROOF · ONE BINARY CHOICE PER ELEMENT

To build a subset $X \subseteq A$, walk the n elements of A and for each decide: in X, or not in X. Two options, n times, independently.

Every subset arises from exactly one such sequence of decisions, so there are $2 \times 2 \times \cdots \times 2 = 2^n$ of them. ■

CHECK IT $|A| = 0$: only $\varnothing$, and $2^0 = 1$. $|A| = 3$: the eight we listed, $2^3 = 8$.

GROWS FAST $|A| = 10$ gives 1024 subsets; $|A| = 20$, over a million. Session 4 names this: exponential.

Products and power sets

ACTIVITY 4 PAIRS 12 MIN

Let $A = \{1, 2\}$ and $B = \{a\}$.

1. List $A \times B$ and $B \times A$. Are they equal? Justify from the definition of an ordered pair.
2. List $\mathcal{P}(A)$ in full. How many elements does $\mathcal{P}(\mathcal{P}(A))$ have? Do not list it.
3. Is $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$? Prove it or break it.
4. Is $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$? Prove it or break it.

Share: one pair presents (3), another (4). The room decides why one direction of (4) survives and the other does not.

Why $\cap$ survives and $\cup$ does not

1 · $A = \{1, 2\}$, $B = \{A\}$

$A \times B = \{(1, a), (2, a)\}$, $B \times A = \{(a, 1), (a, 2)\}$.

Not equal: $(1, a) = (a, 1)$ needs $1 = a$.

2 · POWER SETS GROW $\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$, so $|\mathcal{P}(\mathcal{P}(A))| = 2^4 = 16$.

3 · $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ – TRUE $X \subseteq A \cap B$ means $X \subseteq A$ and $X \subseteq B$ – exactly the right-hand side.

4 · $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ – FALSE With $A = \{1\}$, $B = \{2\}$: $\{1, 2\}$ is a subset of $A \cup B$ but of *neither* alone. Only $\supseteq$ holds.

WATCH OUT

Three mistakes worth avoiding

1 · CONFUSING $\in$ WITH $\subseteq$ Is $\{1\} \in \{\{1\}, 2\}$? Yes. Is $\{1\} \subseteq \{1, 2\}$? Yes. Different relations, different questions.

2 · THE DIAGRAM IS NOT THE PROOF A Venn diagram shows one configuration; the claim quantifies over all sets. Draw to guess, write the inclusion to prove.

3 · FORGETTING $\varnothing$ $\varnothing \subseteq A$ for every A. It is in every power set — and it is the case people drop when listing subsets.

A RELIABLE FIX Go back to the definition and read it as a sentence about an arbitrary element. These mistakes usually come from reasoning from the shape of the notation instead.

On your own, before you leave

1 With $U = \{1, \ldots, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$: compute $A \triangle B$ and $(A \cap B)^c$.

2 Prove $(A \cap B) \subseteq A$ for all sets A, B. Two lines, and the word *arbitrary* must appear.

3 Disprove: $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$. Name your sets and the element that breaks it.

RULE FOR TODAY One sentence of justification under every answer. Hand it in on the way out — not graded, but it tells us who to sit with tomorrow.

WRAP-UP

What today was actually about

→ A set is determined by **what is in it** — not by order, repetition, or the notation you wrote it in.

→ Every operation is a logical connective in disguise: $\cup$ is *or*, $\cap$ is *and*, $\setminus$ is *and not*.

→ **Double inclusion** proves an equality; **one counterexample** destroys one.

set · element · subset · double inclusion · power set · De Morgan · Cartesian product · Venn diagram

Homework 2

6 identities to prove by double inclusion; **3 to disprove** by counterexample. Proofs begin with an arbitrary element. Disproofs name the sets and a separating element.

Next session: **Functions** — which are built from today's Cartesian product, and are the reason ordered pairs mattered.